



KLE Technological University

Creating Value,
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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
KLE TECHNOLOGICAL UNIVERSITY, HUBBALLI, INDIA

Capstone Project Report

On

BrainHive - an AI-Powered Study Group Platform

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IN

COMPUTER SCIENCE AND ENGINEERING

Submitted By

_____	01FE22BCS—
Apoorva	01FE22BCS314
Drushti G P	01FE22BCS177
_____	01FE22BCS

Under the guidance of

Vijay Biradar



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CERTIFICATE

This is to certify that the project entitled **BrainHive - an AI-Powered Study Group Platform** is a bonafide work carried out by the student team ———, Apoorva - 01FE22BCS314, Drushti G P - 01FE22BCS177, ———, in partial fulfillment of the completion of the 8th semester B.E. course during the year 2025–2026. The project report has been approved as it satisfies the academic requirement concerning the project work prescribed for the above-mentioned course.

Guide Name:
Vijay Biradar

Head, Department of CSE
Dr. Vijayalakshmi M.

External Viva-Voce

Name of the examiners

Signature with date

1 _____

2 _____

ABSTRACT

Collaborative learning is widely recognized as an effective approach to enhance student engagement, knowledge retention, and academic performance. However, in practice, students face significant challenges in forming and managing productive study groups due to reliance on multiple disconnected tools such as messaging applications, scheduling platforms, and email systems. This fragmented ecosystem leads to inefficient coordination, communication gaps, and reduced learning effectiveness, particularly in hybrid and remote learning environments. This paper presents **BrainHive — an AI-Powered Study Group Platform**, a unified web-based solution designed to streamline group discovery, management, scheduling, and communication within a single system. The platform is built using modern web technologies, including React for the frontend, Node.js with Express for the backend, and MySQL for data management, ensuring scalability, security, and robust performance. BrainHive enables students to seamlessly create or join study groups, organize structured sessions, and engage in collaborative discussions through an integrated interface. A key contribution of this work is the incorporation of multiple intelligent AI agents that enhance the collaborative learning experience. These include a Study Buddy Agent for real-time academic assistance, a Session Summarizer Agent for generating structured discussion summaries, a Group Recommendation Agent for personalized group suggestions, an Onboarding Agent for user guidance, and a Platform Health Agent for system monitoring and analytics. The proposed system significantly reduces coordination overhead while improving the quality of peer-to-peer interaction and structured learning. By integrating collaboration tools with AI-driven assistance, BrainHive not only enhances learning efficiency but also provides a scalable framework for next-generation academic collaboration platforms.

Keywords : *Collaborative Learning, AI in Education, Study Group Platform, Intelligent Agents, Web-Based Systems, Personalized Learning, EdTech*

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- 01FE22BCS

Apoorva - 01FE22BCS314

Drushti G P - 01FE22BCS177

- 01FE22BCS

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Chapter 1

INTRODUCTION

In today's modern learning environment, education is no longer limited to individual study. Students increasingly benefit from collaborative learning, where they work together, exchange ideas, and support each other in understanding complex concepts. Group study not only improves comprehension but also enhances critical thinking, communication skills, and motivation. When students learn in groups, they are exposed to different perspectives, which helps them approach problems more effectively and gain deeper insights into subjects. Despite these advantages, organizing and maintaining effective study groups remains a challenge. Although many digital tools are available, they are not specifically designed for academic collaboration. Students often rely on a combination of messaging applications, video conferencing tools, calendars, and emails to coordinate their study activities. While each of these tools serves a particular purpose, using multiple platforms creates fragmentation and inefficiency. Important messages may be missed, schedules may conflict, and managing group activities becomes time-consuming and confusing. Another major issue is the difficulty in finding suitable study partners. Students typically form study groups through informal methods such as friends, classmates, or social media. These approaches do not guarantee compatibility in terms of subject interest, learning pace, or availability.

As a result, many groups become inactive, unorganized, or ineffective. Inconsistent participation and poor coordination often lead to missed study sessions and reduced productivity, preventing students from fully benefiting from collaborative learning. With the increasing adoption of online and hybrid learning models, these challenges have become even more significant. Students are no longer limited to interacting within their physical classrooms and often need to collaborate with peers across different locations. However, existing tools do not provide a structured way to support this type of academic interaction. The lack of a unified platform designed specifically for study groups creates a gap in the current educational ecosystem. To address these challenges, this project introduces **Brain Hive – Study Group Finder**, a web-based platform designed to simplify and enhance collaborative learning. The primary goal of Brain Hive is to provide a centralized system where students can easily discover, create, and manage study groups based on their subjects, interests, and availability. By bringing all essential features into a single platform, Brain Hive eliminates the need to switch between multiple tools, thereby improving efficiency and user experience. The platform offers a range of features that support structured and effective group study. Students can create study groups,

join existing ones, and manage memberships with ease. Group leaders can schedule study sessions, set topics, and organize discussions, while members can view upcoming sessions and stay informed through notifications. A built-in communication system enables real-time interaction, allowing students to discuss topics, clarify doubts, and collaborate seamlessly within the platform.

In summary, Brain Hive addresses a common problem faced by students by offering a unified and structured solution for study group management. It simplifies the process of finding study partners, organizing sessions, and communicating effectively. By reducing dependency on multiple tools and enhancing collaboration, the platform enables students to make better use of group learning and improve their overall academic performance.

1.1 Sustainable Development Goals

The proposed **Brain Hive – Study Group Finder** platform contributes directly to multiple United Nations Sustainable Development Goals (SDGs):

- **SDG 4 – Quality Education:** Ensures inclusive and equitable access to learning opportunities by providing a free, web-based platform where students can discover and join study groups regardless of their institution or socioeconomic background. It promotes collaborative learning, improves understanding through peer interaction, and supports structured study practices (Target 4.3 – Ensure equal access to affordable and quality technical, vocational, and tertiary education).
- **SDG 10 – Reduced Inequalities:** Reduces educational disparities by enabling students from smaller or less-resourced institutions to collaborate with peers from larger universities. The platform provides equal opportunities for participation, helping bridge gaps caused by differences in access to academic resources and networks.
- **SDG 17 – Partnerships for the Goals:** Encourages collaboration and knowledge sharing among students across institutions by creating a connected academic community. It fosters partnerships that support mutual learning, idea exchange, and long-term academic and professional relationships.

In addition, Brain Hive strengthens the overall learning ecosystem by promoting lifelong learning habits. It encourages self-directed study, builds communication and teamwork skills, and supports continuous knowledge exchange. These aspects contribute to creating a more engaged, skilled, and collaborative student community aligned with the broader vision of sustainable and inclusive education.

1.2 Motivation

The motivation behind developing **Brain Hive – Study Group Finder** comes from the gap between the known benefits of collaborative learning and the difficulties students face in actually practicing it. Studying in groups has been shown to improve understanding, boost academic performance, and significantly increase retention of concepts. However, in reality, many students struggle to find suitable study partners or maintain consistent group activities. One major reason for this is the reliance on multiple disconnected tools for communication, scheduling, and coordination. Instead of focusing on learning, students often spend a considerable amount of time organizing study sessions, which reduces their overall productivity. This problem has become even more evident with the rise of hybrid and remote learning, where students are no longer limited to physical classrooms but still lack proper platforms to connect with peers who share similar academic goals and schedules. Existing tools are not designed specifically for academic collaboration, making the process inefficient and unstructured. As a result, many potential learning opportunities are lost due to poor coordination and lack of accessibility.

1.3 Literature Survey

This section presents a detailed review of existing tools, platforms, and research works related to collaborative learning, communication systems, and AI-assisted education. Over the years, several technologies have been developed to support online learning and interaction among students. However, most of these systems focus only on specific aspects such as communication or individual learning, rather than providing a complete solution for collaborative academic environments. This section analyzes their contributions, strengths, and limitations, thereby highlighting the need for an integrated platform like BrainHive.

1.3.1 Discord for Active Learning Communities

Fisher et al. [1] explored the use of Discord as a platform to replicate classroom interaction during the COVID-19 pandemic. Discord, originally designed for gaming communities, was adapted for educational purposes due to its support for both synchronous and asynchronous communication. The platform allows users to create servers with multiple channels, enabling organized discussions, group collaboration, and resource sharing. Features such as voice channels, video calls, screen sharing, and real-time messaging helped simulate a classroom-like environment.

The study found that students actively engaged in discussions, participated in peer learning, and felt more comfortable asking questions compared to traditional classroom settings.

Discord also allowed continuous interaction beyond scheduled class hours, improving overall engagement.

However, the platform has several limitations when used for academic purposes. It lacks structured group formation mechanisms, making it difficult to organize students based on subjects or learning goals. Additionally, there are no built-in tools for scheduling study sessions, tracking progress, or providing academic feedback. Managing large groups also becomes challenging without proper moderation tools. These limitations highlight that while Discord is effective for communication, it is not a complete solution for structured academic collaboration.

1.3.2 Peer Learning in Higher Education

Smith et al. [2] conducted a comprehensive review of peer learning in both online and face-to-face educational environments. Peer learning is recognized as an effective method for improving understanding, as students learn by explaining concepts to one another and engaging in discussions.

The study emphasized that online learning environments offer greater flexibility and accessibility, allowing students to collaborate regardless of geographical constraints. On the other hand, face-to-face learning provides stronger interaction, immediate feedback, and better engagement.

Despite these benefits, the study identified several challenges in online peer learning systems. Most platforms lack structured mechanisms for forming groups, assigning roles, and ensuring accountability among members. As a result, participation levels vary, and group effectiveness decreases over time. The absence of proper coordination tools often leads to confusion and reduced productivity. This indicates the need for systems that combine the flexibility of online learning with structured collaboration features.

1.3.3 HELMeTO Framework for E-learning

Johnson et al. [3] proposed the HELMeTO (Helping Learners Manage and Track Online learning) framework, which provides a structured approach to designing e-learning platforms. The framework categorizes essential features into three main components: communication, collaboration, and learning support.

Communication features include messaging, commenting, and discussion forums, while collaboration features involve group activities, shared workspaces, and user interactions. Learning support features include feedback systems, assessments, and progress tracking.

The study found that most existing platforms implement only a subset of these features, forcing users to rely on multiple tools to complete their tasks. For example, a platform may

support communication but lack scheduling or feedback mechanisms. This fragmentation reduces usability and efficiency. The framework highlights the importance of integrating all essential features into a single platform to provide a seamless learning experience.

1.3.4 Slack for Academic Collaboration

Brown et al. [4] examined the use of Slack as a communication and collaboration tool in academic environments, particularly in seminars and project-based courses. Slack provides organized communication through channels, direct messaging, and file sharing, making it suitable for teamwork and discussion.

The study found that students actively used Slack for asking questions, sharing resources, and coordinating group tasks. It improved transparency, as all conversations were visible within channels, and encouraged peer-to-peer learning. Many students benefited from reading discussions even without actively participating.

However, Slack also has limitations when used for academic purposes. It is primarily a communication tool and does not include features such as structured group formation, session scheduling, or academic guidance. Information overload can also occur due to continuous messaging, making it difficult to track important discussions. Additionally, it requires active monitoring by instructors or moderators to ensure effective usage. These limitations show that Slack, while useful for communication, does not fully support structured collaborative learning.

1.3.5 AI in Personalized Learning Systems

Zawacki-Richter et al. [5] conducted a systematic review of Artificial Intelligence applications in higher education. The study focused on how AI can enhance learning through personalization, automation, and intelligent support systems. AI technologies such as recommendation systems, chatbots, and adaptive learning platforms help tailor content based on individual student needs.

The study highlighted that AI can significantly improve learning efficiency by providing instant feedback, suggesting relevant resources, and automating repetitive tasks. Intelligent tutoring systems can guide students step-by-step, improving understanding and performance.

However, one of the key limitations identified is that many AI-based systems focus primarily on individual learning rather than collaborative environments. They are often not integrated with group-based platforms, limiting their effectiveness in supporting peer learning. This creates a gap where AI capabilities are not fully utilized in collaborative educational settings.

1.3.6 Intelligent Tutoring Systems (ITS)

Woolf et al. [6] explored Intelligent Tutoring Systems (ITS), which are designed to provide personalized instruction and feedback to learners. These systems simulate one-on-one tutoring by adapting to the learner's pace, strengths, and weaknesses.

ITS has been shown to significantly improve learning outcomes by offering tailored guidance and continuous feedback. Students can learn independently and receive support without the need for constant instructor involvement.

Despite these advantages, ITS primarily focuses on individual learning and does not support group interaction or collaboration. It lacks features for communication, group coordination, and peer learning, which are essential components of modern education. Therefore, while ITS is effective for personalized learning, it is not sufficient for collaborative learning environments.

1.3.7 Summary and Research Gap

The literature review clearly shows that existing systems address only specific aspects of learning. Platforms like Discord and Slack focus on communication, while AI-based systems and ITS focus on individual learning. However, none of these solutions provide a unified platform that supports structured group formation, scheduling, communication, and intelligent assistance together. Key limitations identified include lack of integration, absence of academic-specific features, poor group coordination, and limited use of AI in collaborative environments. These gaps highlight the need for a comprehensive solution that combines all essential features into a single platform. The proposed system, **BrainHive**, addresses these challenges by integrating group discovery, scheduling, communication, and AI-powered assistance into one unified platform. It not only simplifies collaboration but also enhances the overall learning experience by providing intelligent support and structured interaction among students.

1.4 Problem Statement

To design and develop a scalable and secure web-based platform that enables students to efficiently discover, create, and manage study groups through integrated features such as intelligent search, real-time communication, and session scheduling, while ensuring data privacy and reliable system performance.

1.5 Problem Analysis

Collaborative learning is hindered by the absence of a unified platform, forcing students to use multiple disconnected tools for communication, scheduling, and coordination. This fragmentation leads to inefficient communication, poor organization, and missed study opportunities. Students also struggle to find compatible study partners, resulting in inactive or ineffective groups. Additionally, significant time is wasted on managing logistics instead of focusing on actual learning. The lack of intelligent assistance and structured support further reduces the effectiveness of study groups, highlighting the need for an integrated AI-powered solution.

1.5.1 Design Principles

The design of the proposed **BrainHive – AI-Powered Study Group Platform** is based on fundamental software engineering principles:

- **Modularity:** The system is divided into independent modules such as user management, group management, scheduling, chat system, and AI agents, enabling easier development, testing, and maintenance.
- **Scalability:** The architecture supports increasing users, groups, and AI workloads without significant redesign.
- **Efficiency:** Integration of features into a single platform reduces coordination overhead and improves user productivity.
- **Robustness:** The system is designed to handle real-world issues such as network delays, inactive users, and inconsistent participation.
- **Reliability:** Consistent performance is ensured for scheduling, messaging, and AI responses across different usage conditions.
- **Reusability:** Core components such as chat modules and AI services can be reused or extended for future enhancements.
- **Maintainability:** Clean code structure and documentation allow easy updates and long-term maintenance.
- **Data Integrity:** Accurate synchronization of user data, group information, and session details is maintained.
- **Fault Tolerance:** The system continues to function even if certain services (e.g., AI agents) fail temporarily.

- **User-Centric Design:** Focus on usability and accessibility ensures a smooth and intuitive user experience.
- **Security and Privacy:** Implementation of authentication and secure data handling ensures protection of user information.

1.5.2 Scope

The scope of this project focuses on the development of an AI-powered web platform for collaborative learning. The major components and boundaries are defined as follows:

Included in Scope:

- Study group discovery based on subjects, interests, and availability.
- AI-based group recommendation system.
- Group creation and membership management.
- Session scheduling and reminders.
- Real-time group chat functionality.
- AI agents for study assistance, summarization, onboarding, and system monitoring.
- User authentication and profile management.

Out of Scope:

- Real-time video conferencing implementation.
- Mobile application development.
- Integration with institutional academic systems.
- Advanced analytics for academic performance prediction.

1.5.3 Constraints

The design and implementation of the proposed system are subject to several constraints:

- **Computational Constraints:** AI-based features require additional processing resources, especially for real-time responses.
- **Time Constraints:** Development is limited by academic project timelines.

- **User Engagement Constraints:** Effectiveness depends on active participation by users.
- **Data Privacy Constraints:** Secure handling of user data is essential.
- **Scalability Constraints:** System performance must be maintained as the number of users increases.
- **Network Constraints:** Real-time features depend on stable internet connectivity.

1.6 Objectives

1. To design and develop a web-based platform for efficient study group discovery and management.
2. To implement AI-powered agents that assist in learning, summarization, and recommendations.
3. To provide integrated scheduling and real-time communication features.
4. To reduce coordination effort and improve collaborative learning efficiency.
5. To ensure system scalability, security, and reliability for real-world usage.

Chapter 2

Proposed System

To improve collaborative learning and simplify study group management, the proposed system **BrainHive — AI-Powered Study Group Platform** integrates group discovery, scheduling, communication, and intelligent assistance into a unified web-based platform. Unlike existing fragmented solutions, the system combines core functionalities with AI-powered agents to enhance user experience and learning outcomes. The platform leverages modern web technologies along with intelligent automation to reduce coordination effort and provide personalized academic support.

2.0.1 Description of Proposed System

The proposed system is designed using a modular web-based architecture that separates the application into distinct layers, ensuring better organization, scalability, and maintainability. The frontend is developed using React, which provides a dynamic and responsive user interface, allowing users to interact smoothly with the platform. The backend is implemented using Node.js and Express, which handle API requests, user authentication, and overall application logic. MySQL is used as the database to efficiently store and manage user information, study group details, chat messages, and scheduled session data. The system workflow begins with user registration and authentication, where students create their profiles and specify their field of study. This information plays a key role in personalizing the user experience. Based on the profile data, an AI-based recommendation engine analyzes user preferences and activity to suggest relevant study groups. This helps students find suitable peers without relying on informal methods. Users can either create new study groups or join existing ones. Group leaders are provided with administrative controls to manage membership requests, organize group activities, and maintain coordination.

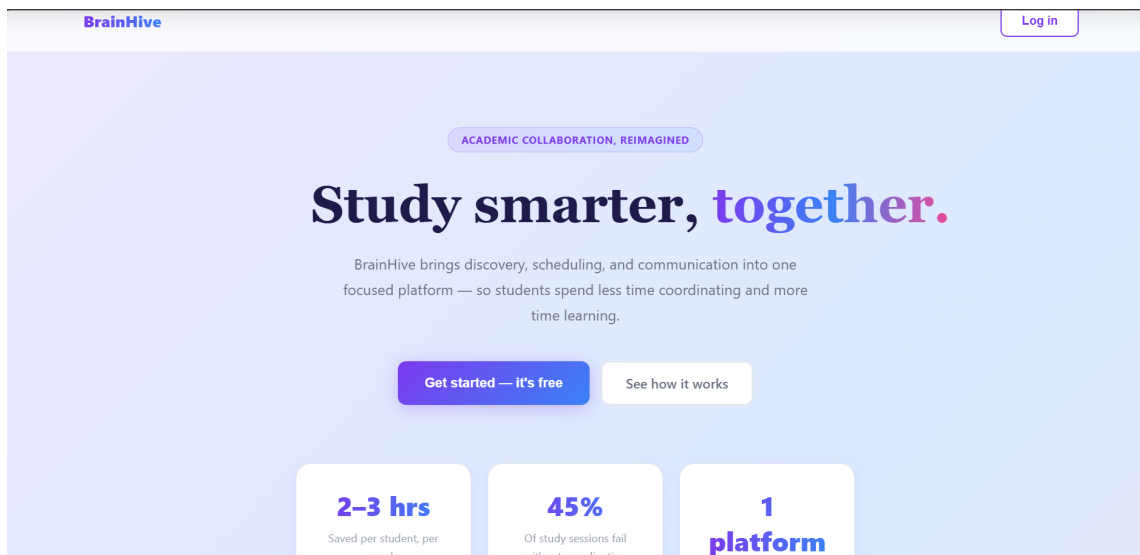


Figure 2.1: BrainHive Dashboard – Group Discovery and Management Interface

As shown in Figure 2.1, the dashboard provides a centralized view where users can explore study groups, manage memberships, and access important updates. This interface simplifies navigation and ensures that all essential features are easily accessible in one place. Each study group includes a real-time chat system that allows members to communicate and discuss topics. Messages are updated at regular intervals to provide near real-time interaction. The chat system is enhanced with an AI-powered Study Buddy Agent that responds to academic queries instantly, improving learning efficiency.

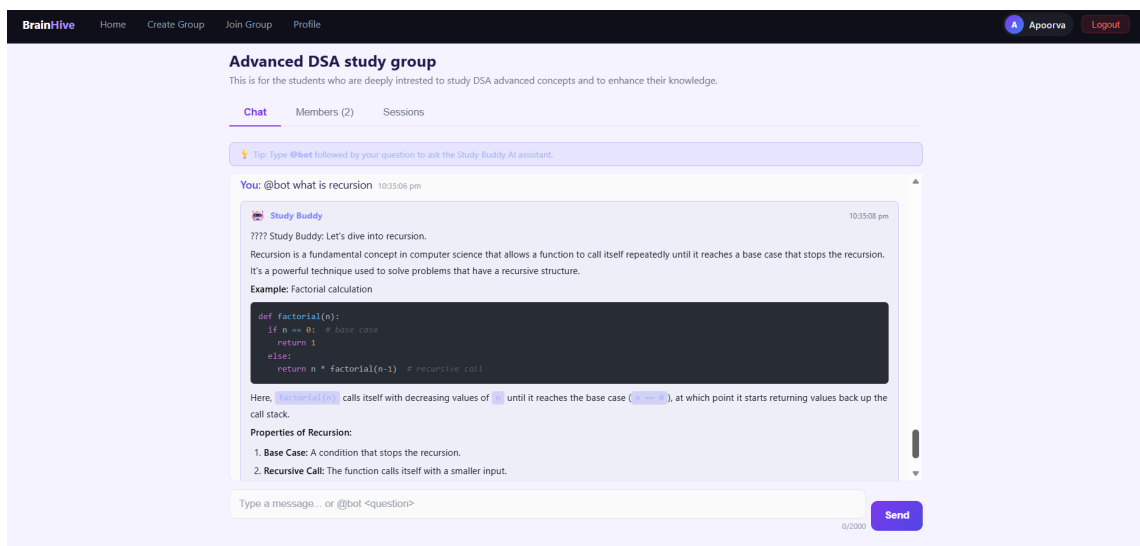


Figure 2.2: AI Chat Interface – Study Buddy Agent Providing Academic Assistance

Figure 2.2 illustrates the AI chat interface where users can interact with the Study Buddy Agent by typing queries. The agent provides explanations, examples, and clarifications, making

ing the discussion more interactive and informative. The session scheduling module allows group leaders to create study sessions with defined topics and timings, while members can view upcoming sessions and receive updates. Additionally, the Session Summarizer Agent generates structured summaries of discussions, helping users quickly review key points.

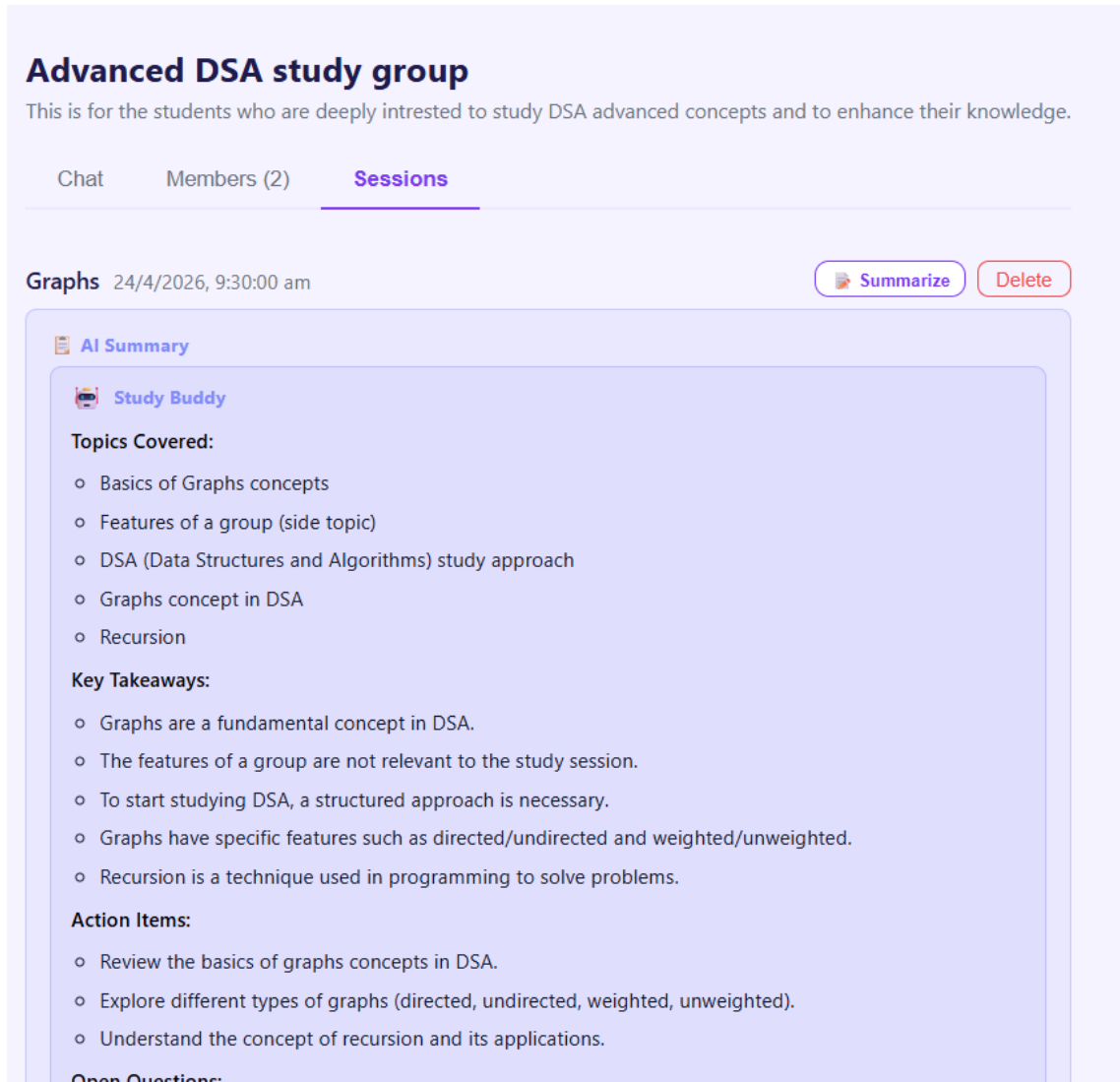
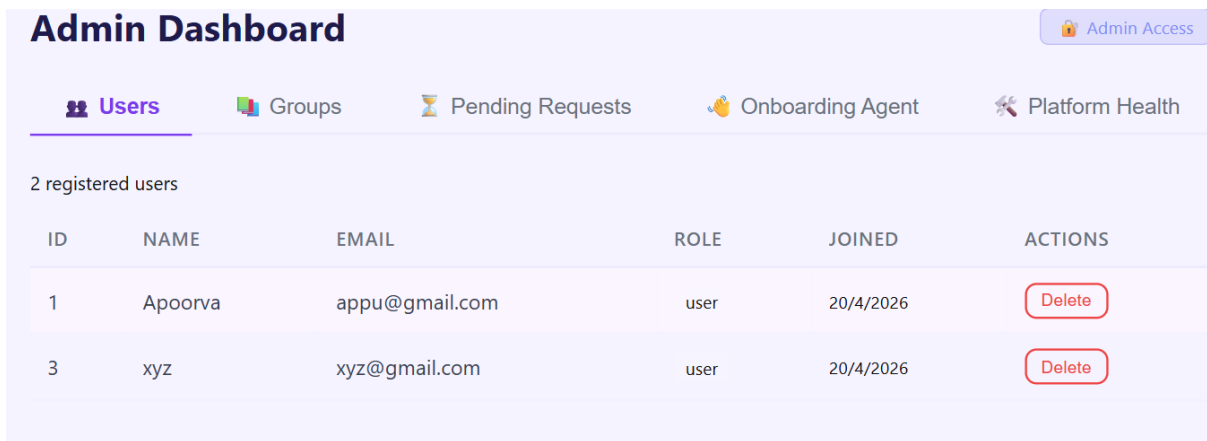


Figure 2.3: Session Management and AI Summarizer Output

As shown in Figure 2.3, users can view scheduled sessions along with AI-generated summaries that highlight important topics, key takeaways, and action items. This feature improves knowledge retention and ensures effective revision. In addition to student-facing features, the system includes an **Admin Dashboard** that provides centralized control over platform operations. Administrators can monitor registered users, manage study groups, and handle pending membership requests. The dashboard also includes sections for onboarding and platform health monitoring, enabling efficient system management.



The image shows a screenshot of an 'Admin Dashboard' with a light purple header and a white main content area. The header includes the title 'Admin Dashboard' and a button labeled 'Admin Access'. Below the header, there are five navigation tabs: 'Users' (selected), 'Groups', 'Pending Requests', 'Onboarding Agent', and 'Platform Health'. The 'Users' tab displays '2 registered users' and a table with columns: ID, NAME, EMAIL, ROLE, JOINED, and ACTIONS. The table lists two users: ID 1, Name Apoorva, Email appu@gmail.com, Role user, Joined 20/4/2026, and a 'Delete' button; and ID 3, Name xyz, Email xyz@gmail.com, Role user, Joined 20/4/2026, and a 'Delete' button.

ID	NAME	EMAIL	ROLE	JOINED	ACTIONS
1	Apoorva	appu@gmail.com	user	20/4/2026	Delete
3	xyz	xyz@gmail.com	user	20/4/2026	Delete

Figure 2.4: Admin Dashboard – User Management and Platform Monitoring

As shown in Figure 2.4, administrators can view user details such as ID, name, email, role, and join date, and perform actions like removing users if necessary. The dashboard also provides insights into system activity and supports AI-driven monitoring to detect inactive groups or anomalies. This ensures better control, improved reliability, and smooth functioning of the platform.

A key feature of the system is the integration of five AI agents that enhance functionality. The Study Buddy Agent provides instant responses to academic queries within the chat. The Session Summarizer Agent generates structured summaries from group discussions. The Group Recommendation Agent suggests suitable groups based on user profiles and activity. The Onboarding Agent assists new members by providing relevant group information, and the Platform Health Agent helps administrators monitor system performance and detect issues. Security and reliability are ensured through JWT-based authentication, encrypted password storage, and rate limiting. The system is designed to be scalable and maintainable, allowing future extensions and integration of additional AI capabilities.

2.0.2 Application of Proposed System

- **Academic Collaboration:** Enables students to form effective study groups, share knowledge, and improve understanding through peer-to-peer learning.
- **Remote and Hybrid Learning:** Supports collaboration across different locations, making it suitable for online and hybrid education environments.
- **Exam Preparation:** Helps students organize structured study sessions and revise topics efficiently with AI-generated summaries.
- **Skill Development:** Encourages teamwork, communication, and problem-solving skills essential for academic and professional growth.

- **Institutional Use:** Can be adapted by colleges and universities to enhance student engagement and collaborative learning practices.

2.0.3 Advantages of Proposed System

- **Unified Platform:** Integrates group discovery, scheduling, communication, and AI assistance into a single system, reducing dependency on multiple tools.
- **AI-Driven Learning Support:** Provides real-time academic assistance, automated summaries, and personalized recommendations.
- **Improved Efficiency:** Reduces time spent on coordination and increases focus on actual learning activities.
- **Enhanced Collaboration:** Enables structured and effective group interactions with better organization and communication.
- **Scalability and Flexibility:** Supports a growing number of users and allows integration of additional features and AI services.
- **User-Friendly Design:** Simple and intuitive interface ensures ease of use for students across different disciplines.

Chapter 3

IMPLEMENTATION

The implementation of **BrainHive — AI-Powered Study Group Platform** follows a modular full-stack architecture consisting of frontend, backend, database, and AI service integration. The system is designed with a strong focus on scalability, security, and maintainability, ensuring that it can support a growing number of users and evolving feature requirements. Each layer of the system is developed independently while maintaining seamless communication through well-defined APIs. This modular approach simplifies debugging, testing, and future enhancements.

3.0.1 Frontend Implementation

The frontend is developed using **React**, which enables the creation of a dynamic and interactive user interface. To efficiently manage application-wide state, **Redux Toolkit** is used, allowing predictable and centralized state handling. Navigation across different pages such as dashboard, chat, sessions, and profile is handled using **React Router**, ensuring a smooth user experience. The user interface is designed to be intuitive, responsive, and accessible across devices. Key components such as group dashboards, chat interfaces, session schedulers, and user profiles are implemented as reusable and modular components, improving code maintainability and scalability. The chat system uses periodic polling every 4 seconds to simulate near real-time communication, ensuring users receive timely updates without complex real-time infrastructure.

Additionally, **React Markdown** is integrated to properly render AI-generated responses, including formatted text, code snippets, and structured outputs. This enhances readability and improves the overall interaction experience with AI agents. Proper error handling, loading states, and UI feedback mechanisms are also implemented to ensure a smooth and user-friendly experience.

3.0.2 Backend Implementation

The backend is implemented using **Node.js** and **Express**, providing a robust and scalable server-side environment. The backend is responsible for handling all API requests, managing business logic, and ensuring secure communication between the frontend and database. RESTful APIs are designed for core functionalities such as user management, group creation, session

scheduling, and chat handling. Each API endpoint follows a structured approach with proper validation, error handling, and response formatting. Authentication is implemented using **JWT (JSON Web Tokens)**, enabling secure and stateless user sessions. User passwords are securely stored using **bcrypt** hashing, ensuring protection against unauthorized access. Middleware functions are used for request validation, authentication checks, and logging. Additionally, **rate limiting** is implemented to prevent abuse of APIs and ensure system stability under high traffic conditions. The backend is structured in a modular way, separating routes, controllers, and services to improve maintainability.

3.0.3 Database Implementation

The system uses **MySQL** as the primary database for storing all application data. A well-structured relational schema is designed to maintain consistency and integrity of data. Key tables include users, groups, memberships, chat messages, and session schedules, each connected through defined relationships using foreign keys. Efficient querying techniques such as indexing and optimized joins are used to improve performance, especially for frequently accessed data like group listings and chat messages. Auto-migration strategies are implemented to manage database schema updates, ensuring smooth transitions during development and deployment. Data validation is enforced at both application and database levels to prevent inconsistencies. Backup and recovery strategies can also be integrated to ensure data reliability in production environments.

3.0.4 AI Agent Integration

BrainHive integrates five AI agents using a flexible and pluggable provider system that supports APIs such as OpenAI, Hugging Face, and local inference engines like vLLM. This design allows easy switching or upgrading of AI models without affecting the core system.

Each AI agent is designed as an independent service that handles specific tasks:

- The **Study Buddy Agent** processes user queries from the chat interface and generates contextual explanations, examples, and clarifications in real time.
- The **Session Summarizer Agent** analyzes chat transcripts and produces structured summaries, including key points, action items, and important discussions.
- The **Group Recommendation Agent** uses user profile data, interests, and activity patterns to suggest relevant study groups, improving discovery and engagement.
- The **Onboarding Agent** automatically sends personalized welcome messages to new users, guiding them through platform features and upcoming sessions.

- The **Platform Health Agent** monitors system performance, identifies inactive groups or anomalies, and generates alerts for administrators.

This AI integration significantly enhances the functionality of the platform by automating tasks, improving learning efficiency, and providing intelligent support.

3.0.5 System Workflow

The system workflow begins with user registration and authentication. After logging in, users complete their profiles by specifying their field of study and preferences. Based on this information, the recommendation agent suggests suitable study groups. Users can either join existing groups or create new ones. Once part of a group, they can participate in discussions through the chat system, schedule study sessions, and collaborate with other members. Group leaders manage requests and organize activities efficiently. AI agents operate continuously in the background, providing real-time assistance, generating summaries, and improving recommendations. This ensures a seamless and intelligent user experience where students can focus more on learning rather than coordination.

3.0.6 Security and Performance Considerations

Security is a critical aspect of the system. It is ensured through JWT-based authentication, encrypted password storage using bcrypt, and strict API access control. Input validation and sanitization are implemented to prevent common vulnerabilities such as SQL injection and cross-site scripting. Rate limiting is applied to control excessive requests and protect the system from abuse. Performance optimization techniques include efficient database queries, caching strategies (if extended), and lightweight frontend rendering. The modular architecture allows the system to scale horizontally, supporting multiple users simultaneously without performance degradation. Overall, the platform is designed to deliver a reliable, secure, and efficient experience for collaborative learning.

Chapter 4

RESULTS & DISCUSSIONS

The implementation of **BrainHive — AI-Powered Study Group Platform** demonstrates a significant improvement in the way students collaborate and manage study activities. The system was tested with multiple users forming groups, scheduling sessions, and interacting through the platform to evaluate its effectiveness and usability. The results show that integrating all functionalities into a single platform greatly reduces the time spent on coordination. Users were able to create and join study groups easily, and the centralized system eliminated the need to switch between multiple tools. The real-time chat feature enabled smooth communication among group members, ensuring active participation and continuous discussion.

The AI agents played a key role in enhancing the overall learning experience. The Study Buddy Agent provided instant responses to academic queries, helping users clarify doubts quickly. The Session Summarizer Agent successfully generated structured summaries from chat discussions, allowing users to review important points without going through entire conversations. The Group Recommendation Agent effectively suggested relevant study groups based on user profiles, improving the chances of forming compatible and active groups. From a system performance perspective, the platform handled multiple users and concurrent operations efficiently. The backend APIs responded reliably, and the database maintained data consistency across group activities, sessions, and messages. Security features such as authentication and data protection mechanisms ensured safe user interactions.

However, certain limitations were observed during testing. The chat system relies on polling, which may introduce slight delays compared to real-time socket-based communication. Additionally, the accuracy of AI-generated responses depends on the underlying model and input quality, which may vary in complex scenarios. Overall, the results indicate that BrainHive successfully addresses the challenges of collaborative learning by providing a structured, efficient, and intelligent platform. The integration of AI-driven features not only improves user engagement but also enhances the quality of learning. Future improvements can focus on optimizing real-time communication, enhancing AI accuracy, and expanding feature capabilities to further strengthen the system.

Chapter 5

CONCLUSION AND FUTURE SCOPE

5.1 Conclusion

The **BrainHive — AI-Powered Study Group Platform** was developed to address a common yet often overlooked challenge in the academic environment: the difficulty students face in forming and managing effective study groups. While collaborative learning has proven benefits in improving understanding, retention, and overall academic performance, the lack of a dedicated and unified platform makes it difficult for students to fully utilize these advantages. BrainHive bridges this gap by providing a structured, centralized system that simplifies group discovery, coordination, and interaction. The platform successfully integrates essential functionalities such as group management, session scheduling, and real-time communication into a single system. By eliminating the need for multiple disconnected tools, it reduces the time and effort students spend on coordination and allows them to focus more on actual learning. The system's user-friendly interface ensures that students can easily navigate and utilize its features without complexity, making it accessible to users from different academic backgrounds. A key contribution of BrainHive is the integration of AI-powered agents that enhance the overall learning experience. Features such as instant academic assistance through the Study Buddy Agent, automated session summaries, and personalized group recommendations provide meaningful support to students. These intelligent capabilities not only improve engagement but also make learning more efficient and interactive. The platform also assists administrators through system monitoring and diagnostics, ensuring smooth and reliable operation. From a technical perspective, the system demonstrates scalability, security, and modularity. The use of modern web technologies such as React, Node.js, and MySQL ensures efficient performance and maintainability. Security mechanisms such as authentication and data protection help safeguard user information, while the modular architecture allows for future enhancements and integration of additional features. The results of the project indicate that BrainHive effectively improves collaboration among students by providing a structured and intelligent environment for group learning. It reduces coordination challenges, enhances communication, and supports consistent participation in study activities. Although certain limitations exist, such as reliance on polling for chat updates and dependency on AI response quality, the overall system performance meets the intended objectives.

5.2 Future Work

While the current system provides a strong foundation, several enhancements can be made to further improve its functionality and impact. One important extension is the implementation of real-time communication using WebSockets to replace polling, enabling faster and more interactive chat experiences. The AI capabilities can be further enhanced by incorporating advanced recommendation algorithms, adaptive learning systems, and personalized study plans based on user performance and behavior. Integration of voice and video conferencing features would allow more interactive and flexible study sessions. Additionally, developing a mobile application can improve accessibility and user engagement. Integration with institutional systems such as Learning Management Systems (LMS) can also expand the platform's usability in academic environments. Future work can also focus on adding analytics dashboards for tracking user progress, participation levels, and group performance. Improving the accuracy and contextual understanding of AI agents will further enhance the quality of assistance provided. These improvements will help transform BrainHive into a more comprehensive and intelligent learning ecosystem, supporting students in achieving better academic outcomes.

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